

Conditions for eigenvalue configurations of two real symmetric matrices: a stochastic definition

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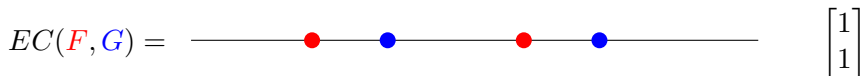
Problem

Definition (Eigenvalue Configuration)

Let $F \in \mathbb{R}^{m \times m}$, $G \in \mathbb{R}^{n \times n}$ be symmetric.

Then their **eigenvalue configuration**, which we write as $EC(F, G)$, is the “relative locations” of the eigenvalues.

Example 1: Let $F = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$ and $G = \begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix}$.



Example 2: Let $F = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$ and $G = \begin{bmatrix} 0 & -1 \\ -1 & 4 \end{bmatrix}$



Problem

Problem

In : $\mathbf{c}$, a desired eigenvalue configuration

Out: Condition on $\mathbf{F} = [a_{ij}]$ and $\mathbf{G} = [b_{ij}]$ such that $\text{EC}(\mathbf{F}, \mathbf{G}) = \mathbf{c}$

Example:

In : $\mathbf{c} =$  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Out: $\begin{pmatrix} \sigma(D_{00}) \\ \sigma(D_{01}) \\ \sigma(D_{10}) \\ \sigma(D_{11}) \end{pmatrix} \in \left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \end{pmatrix} \right\}$ where

$$D_{00} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$D_{01} = \begin{bmatrix} -a_{2,2} - a_{1,1} + 2b_{1,1} & 2b_{1,2} \\ 2b_{1,2} & -a_{1,1} - a_{1,1} + 2b_{2,2} \end{bmatrix}$$

$$D_{11} = \begin{bmatrix} a_{1,1}a_{2,2} - a_{1,2}^2 - (a_{1,1} + a_{2,2})b_{1,1} + b_{1,1}^2 + b_{1,2}^2 & -b_{1,2}(a_{1,1} + a_{2,2} - b_{1,1} - b_{2,2}) \\ -b_{1,2}(a_{1,1} + a_{2,2} - b_{1,1} - b_{2,2}) & a_{1,1}a_{2,2} - a_{1,2}^2 - (a_{1,1} + a_{2,2})b_{2,2} + b_{1,2}^2 + b_{2,2}^2 \end{bmatrix}$$

$\vdots$

Problem: Background and Motivation

Generalization of Descartes' rule of signs: $([1, 2])$

$$\textcolor{red}{F} = [0] \in \mathbb{R}^{1 \times 1} \quad \text{and} \quad \textcolor{blue}{G} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

$$\textcolor{blue}{g}(x) = \det(xI - \textcolor{blue}{G}) = (x + 1)(x - 3)$$



Descartes' rule of signs:

$$\begin{aligned} \textcolor{blue}{g}(x) &= \underbrace{x^2}_{+} \underbrace{-2x}_{-} \underbrace{-3}_{-} \\ \implies v(+, -, -) &= 1 \\ &= \# \text{ positive roots of } \textcolor{blue}{g} \\ &= \text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}). \end{aligned}$$

Problem: Background and Motivation

Applications in Science and Engineering: Many nontrivial problems in science/engineering can be reduced to this problem.

Example (network analysis [3]):

- Let F be the adjacency matrix of a graph.
- Some properties are related to the eigenvalues of F .
- How do eigenvalues change when edges are added or removed?
- Changing edges corresponds to adding a matrix U to F
- Hence, one might be interested in the eigenvalue configuration of

$$F \quad \text{and} \quad G = F + U.$$

Problem

Non-triviality

Why is this problem nontrivial?

Nontrivial specialized quantifier elimination problem.

Problem (Rephrased)

In : $\exists_{\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n} \Phi_{\mathbf{c}}(\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n, a_{ij}, b_{ij})$

Out: $\Psi_{\mathbf{c}}(a_{ij}, b_{ij})$ equivalent to the above.

- Tarski [4] gave first algorithm for quantifier elimination of general formulas over real closed fields.
- Improvements: Collins, Hong, McCallum, Grigorev, Roy, Renegar, Canny, Brown, Strzebonski, Safey-Eldin, ..., Chen [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21]

Q: Why not just use existing algorithms for quantifier elimination?

A: General algorithms not good for specialized input.

Challenge: Develop an algorithm that exploits the problem structure.

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Generic Case: Setup

- Let $F = [a_{ij}] \in \mathbb{R}^{m \times m}$ and $G = [b_{ij}] \in \mathbb{R}^{n \times n}$ be symmetric.
- For now: let F and G be **generic**; i.e. they don't share eigenvalues.
- Label the eigenvalues $\alpha_1 \leq \cdots \leq \alpha_m$ and $\beta_1 \leq \cdots \leq \beta_n$, respectively.
- Let f denote the characteristic polynomial of F .

Generic Case: Definition

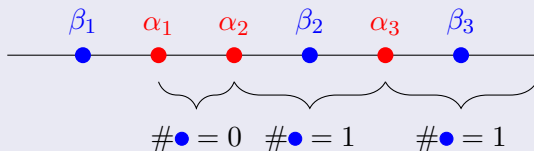
Definition (Generic)

Let $F \in \mathbb{R}^{m \times m}$, $G \in \mathbb{R}^{n \times n}$ be symmetric and generic (i.e. having **distinct eigenvalues**).

Then the **generic** eigenvalue configuration $\text{GEC}(F, G)$ is the vector $\mathbf{c} \in \mathbb{N}^{m \times m}$ where

$$c_t = \#\{j : \beta_j \in (\alpha_t, \alpha_{t+1})\}.$$

Example



$$\text{GEC}(F, G) = (0, 1, 1).$$

Generic Case: Theorem Statement

Theorem

We have $\mathbf{c} = \text{GEC}(\mathbf{F}, \mathbf{G}) \iff \mathbf{c} = C_{\text{sig}} A_{\text{sig}}(\mathbf{F}, \mathbf{G})$, where

① $C_{\text{sig}} = V H^{-1} \in \mathbb{N}^{m \times 2^m}$, where

$$\begin{aligned} V_{t,s} &= \delta_{\{v(s,+), m-t\}} & t &\in \{1, \dots, m\}, \quad s \in \{-, +\}^m \\ H_{e,s} &= s_1^{e_1} \cdots s_m^{e_m} & e &\in \{0, 1\}^m, \quad s \in \{-, +\}^m \end{aligned}$$

② $A_{\text{sig}}(\mathbf{F}, \mathbf{G}) \in \mathbb{Z}^{2^m}$, where

$$\begin{aligned} (A_{\text{sig}})_e &= \sigma(f_e(\mathbf{G})) & e &\in \{0, 1\}^m \\ f_e &= f^{(0)e_0} \cdots f^{(m-1)e_{m-1}} & f^{(k)} &= k\text{-th derivative} \\ f &= \det(xI_m - \mathbf{F}). \end{aligned}$$

Remark: Since all the eigenvalues are real, one can easily express the signature in terms of sign variation count and the entries of $\mathbf{F}$ and $\mathbf{G}$.

Generic Case: Proof Outline

Sketch of Proof: (with full details: ≈ 24 pages in [22])

$$\begin{aligned} \mathbf{c} &= \text{GEC}(F, G) \\ &\Updownarrow \\ \mathbf{c}_t &= \#\{j : \alpha_t < \beta_j < \alpha_{t+1}\} \\ &\Updownarrow \\ \mathbf{c}_t &= \#\{j : \overline{\#\{x : f(x) = 0 \wedge x > \beta_j\}} = m - t\} \\ &\Updownarrow \\ \mathbf{c} &= V \begin{bmatrix} \#\{j : \text{sign}(f^{(0)}(\beta_j), \dots, f^{(m-1)}(\beta_j)) = - \dots -\} \\ \vdots \end{bmatrix} \\ &\Updownarrow \\ \mathbf{c} &= V H^{-1} \begin{bmatrix} \sigma \left((f^{(0)})^0 \dots (f^{(m-1)})^0 (G) \right) \\ \vdots \end{bmatrix} \\ &\Updownarrow \\ \mathbf{c} &= C_{\text{sig}} A_{\text{sig}}(F, G). \end{aligned}$$

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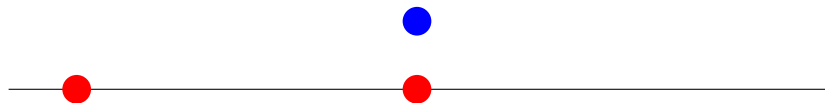
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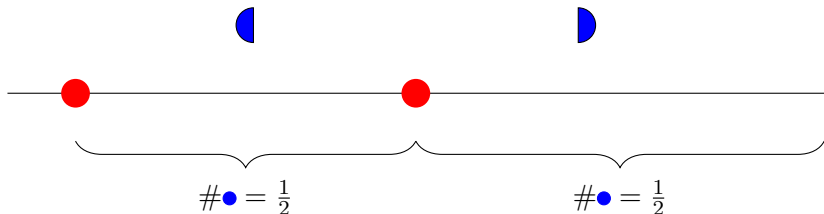
Shared Eigenvalues: Idea



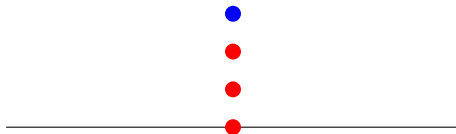
Q: What do we get when we run the algorithm?

A:

$$\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$



Shared Eigenvalues: Idea



$$\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) = \begin{bmatrix} 3/8 \\ 3/8 \\ 1/8 \end{bmatrix}$$

Shared Eigenvalues: New Definition

Definition (Generalized Eigenvalue Configuration)

Let F and G be arbitrary real symmetric matrices. Then

$$\begin{aligned}\text{EC}(F, G) &= \mathbb{E} \text{GEC}(F_d, G) \\ &= \frac{1}{2^m} \sum_{d \in \{-\varepsilon, \varepsilon\}^m} \text{GEC}(F_d, G)\end{aligned}$$

where

$d \in \{-\varepsilon, \varepsilon\}^m$ i.i.d. random variables

F_d = matrix with eigenvalues $\alpha_i + d_i$

ε = "small enough" positive number.

Shared Eigenvalues: New Definition

Definition

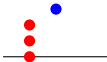
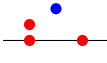
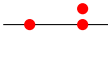
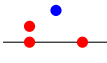
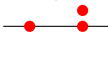
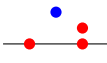
$$\begin{aligned}\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) &= \mathbb{E} \text{GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G}) \\ &= \frac{1}{2^{\textcolor{red}{m}}} \sum_{d \in \{-\varepsilon, \varepsilon\}^{\textcolor{red}{m}}} \text{GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G})\end{aligned}$$

Some remarks:

- ε chosen so that $\textcolor{red}{F}_d$ and $\textcolor{blue}{G}$ do not share eigenvalues.
- ε chosen so that perturbed red points do not move too far.
- New definition still works when $\textcolor{red}{F}$ and $\textcolor{blue}{G}$ do not share eigenvalues.

Shared Eigenvalues: Example



d	c	EC	d	c	EC
— — —		$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	+ — —		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
— — +		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	+ — +		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
— + —		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	+ + —		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
— + +		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$	+ + +		$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) = \frac{1}{8} \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 3/8 \\ 3/8 \\ 1/8 \end{bmatrix}$$

Shared Eigenvalues: Proof Outline

Sketch of Proof: (with full details: ≈ 11 pages in [22])

$$\begin{array}{ccc} \text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) & \overset{?}{=} & C_{\text{sig}} A_{\text{sig}}(\textcolor{red}{F}, \textcolor{blue}{G}) \\ \text{decompose} \quad \parallel & & \parallel \quad \text{combine} \\ \sum_{j=1}^n \text{EC}(\textcolor{red}{F}, [\textcolor{blue}{\beta}_j]) & = & \sum_{j=1}^n C_{\text{sig}} A_{\text{sig}}(\textcolor{red}{F}, [\textcolor{blue}{\beta}_j]) \end{array}$$

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Future Work

Short-term:

- Find applications
- Prune redundancies from output

Long-term:

- Generalize to > 2 matrices
- Generalize to tensors

Other solutions:

- We have another algorithm for GEC which uses the theory of symmetric polynomials [23].
- Can we find more?

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Thank you!